

HAT-P-36b

R-0941

Benchmark run · workflow W8-benchmark-deep · record deep - bench_HATP-36_230312 · created 2026-07-29T13:00:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460015.7723 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.045
Transit depth (Rp/Rs)^2	4.3e-06 +/- 0.0019 [%]
Orbital Inclination (inc)	81.27 +/- 2.75
Airmass coefficient 1 (a1)	14.03 +/- 7.52
Airmass coefficient 2 (a2)	-2.26 +/- 0.89
Scatter in the residuals of the lightcurve fit is	61.15 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	4.0
Transit Duration (day)	0.064 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.045 vs 0.126 (deviation 2.8 σ)
Duration fit vs published	0.064 d vs 0.0925 d (deviation 1.36 σ)
Mid-time O-C	-5.0 min
Depth SNR	0.0
Residual scatter	61.15 %

Light curve

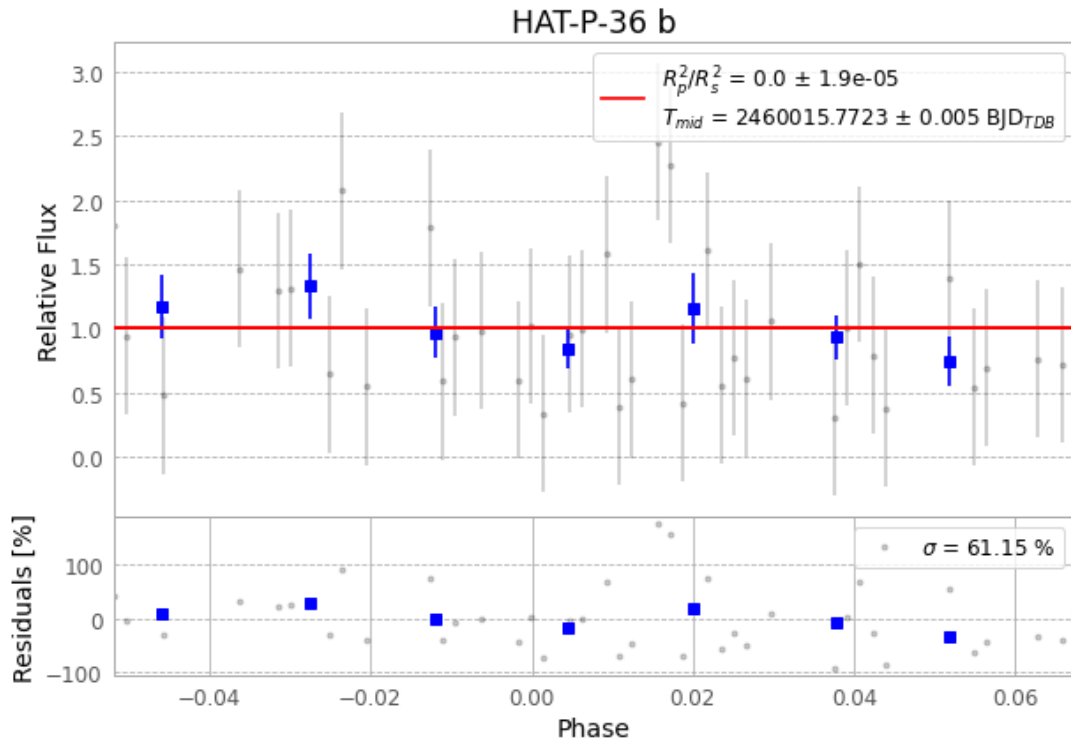


Figure 1 — FinalLightCurve_HAT-P-36 b_2023-03-11.png (SHA-256 b2160483430bea94...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [477, 174], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 54d482dedffb30c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

76 FITS frames (50 MB), HATP-36230312044515.FITS → HATP-36230312083315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-230312.log (cd076f414260...), FinalLightCurve_HAT-P-36 b_2023-03-11.png (b2160483430b...), FinalLightCurve_HAT-P-36 b_2023-03-11.csv (d6610797a2e7...)

Compute resources

Wall time 170.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 13:00Z.